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# Measles Update — United States, January 1–April 17, 2025

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## Abstract

A multistate measles outbreak, predominantly affecting members of close-knit communities with low measles vaccination coverage in New Mexico, Oklahoma, and Texas began in January 2025. As of April 17, a total of 800 cases have been reported in the United States in 2025; 654 (82%) cases in New Mexico, Oklahoma, and Texas have been associated with the ongoing outbreak. These cases represent an approximately 180% increase over the 285 measles cases reported in the United States during all of 2024, and the second highest annual case count in the United States in 25 years. Overall, 771 (96%) patients have been unvaccinated or had unknown vaccination status (77% were unvaccinated, and 14% had unknown vaccination status when excluding 590 cases reported by Texas, which requires explicit consent by law [i.e., opt-in] to enroll in the Texas Immunization Registry), 85 (11%) patients have been hospitalized, and three patients have died. Among 48 (6%) internationally imported cases, 44 (92%) occurred among U.S. residents. Endemic measles was declared eliminated in the United States in 2000 as a direct result of high 2-dose childhood coverage with the measles, mumps, and rubella (MMR) vaccine. However, measles cases and outbreaks continue to occur when travelers with measles return to the United States while they are infectious; larger U.S. outbreaks typically follow importation into close-knit communities with low vaccination coverage. Nationally, risk for widespread measles transmission remains low because of high population-level immunity. To prepare for and prevent measles cases and outbreaks, public health departments should continue working with trusted community messengers on culturally competent community engagement, education, vaccination efforts, and other community infection prevention approaches (e.g., case isolation, contact monitoring, and post-exposure prophylaxis) and coordinating with health care facilities and schools. Increasing national and local MMR vaccination coverage is essential to preventing measles cases and outbreaks.

## Introduction

Measles is the most highly contagious febrile rash illness, infecting up to 90% of susceptible close contacts and resulting in serious complications such as pneumonia, encephalitis, and death. Among the 4,056 measles cases reported in the United States during 2001–2022, a total of 727 (18%) were hospitalized, and three deaths were reported*; of the 727 hospitalized patients, 473 (65%) were unvaccinated, and 187 (26%) had unknown vaccination status (*1*). Worldwide, measles vaccination is estimated to have saved 93.7 million lives during 1974–2024 and played a substantial role in reducing childhood

*Two measles deaths were reported in 2003 (one in a child aged 13 years who had chronic granulomatous disease and one in an adult aged 75 years), and one was reported in 2015 in an adult with immunocompromise aged 28 years.

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mortality (*2*) by preventing complications associated with measles and deaths from other infectious diseases as a consequence of measles-related immunosuppression (*3*).

Endemic measles transmission was declared eliminated[†] in the United States in 2000 after a change from a 1-dose to a 2-dose measles, mumps, and rubella (MMR) vaccination schedule in 1989 (*4*). However, a recent resurgence in global measles, resulting from COVID-19 pandemic–related challenges in implementing measles vaccination routine services and campaigns, has increased the risk for imported cases and outbreaks in the United States, particularly when U.S. travelers are exposed to measles abroad and return to the United States while they are infectious (*5*). Although the United States still benefits from high population immunity from routine MMR vaccination, declining immunization rates among school-aged children and communities with already low vaccination coverage threaten a resurgence of measles, along with its potentially serious associated complications. For this report, CDC used national surveillance data to describe the epidemiology of measles cases and outbreaks reported in the United States during the first 16 weeks of 2025.

[†] Measles elimination is defined as the absence of endemic measles transmission for ≥12 months in the presence of an adequate surveillance system.

## Methods

### Data Source and Case Classification

State health departments notify CDC of confirmed measles cases[§] (*6*) through the National Notifiable Diseases Surveillance System and directly (by email or telephone) to the National Center for Immunization and Respiratory Diseases. Measles vaccination status is ascertained by health departments during each case investigation; patients with written or electronic documentation of receipt of ≥1 dose of a measles-containing vaccine ≥14 days before rash onset are considered vaccinated, and all other patients are classified as unvaccinated or as having unknown measles vaccination status.[¶] Measles cases are classified by the Council of State and Territorial Epidemiologists as internationally imported if 1) at least part of the exposure period (7–21 days before rash onset) occurred outside the United States, 2) rash onset occurred within 21 days of entering the United States, and 3) no known exposure to measles

[§] An acute febrile rash illness with laboratory confirmation (detection of measles virus–specific nucleic acid from a clinical specimen using real-time reverse transcription–polymerase chain reaction or a positive serologic test for measles immunoglobulin M antibody) or direct epidemiologic linkage to a laboratory-confirmed case.

[¶] For residents of Texas, vaccination history is verified in the Texas Immunization Registry (ImmTrac2) or by review of vaccination records; patients with no vaccination records in the registry were considered to have an unverified vaccination history. Texas only disaggregates unvaccinated and unknown vaccination status among hospitalized patients; these records are provider-verified.

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occurred in the United States. All other cases are classified as U.S.-acquired (*6*). For this analysis, patients with imported measles cases were classified as age-eligible for vaccination if they were aged ≥6 months and were not vaccinated according to Advisory Committee on Immunization Practices (ACIP) recommendations (*4*).

### Analysis of Outbreaks

A measles outbreak was defined as the occurrence of three or more epidemiologically linked** cases. Unique measles virus sequences are defined as those differing by at least one nucleotide in the N-450 sequence (i.e., the 450 nucleotides encoding the carboxyl-terminal 150 nucleoprotein amino acids) based on standard World Health Organization recommendations for describing sequence variants†† (*7*). Patients with confirmed vaccine reactions (i.e., rash caused by a reaction to vaccine strain virus) were not included as persons with measles cases, as studies have found no confirmed instances of human-to-human transmission of the measles vaccine strain virus (*6*). This activity was reviewed by CDC, deemed not research, and was conducted consistent with applicable federal law and CDC policy.§§

## Results

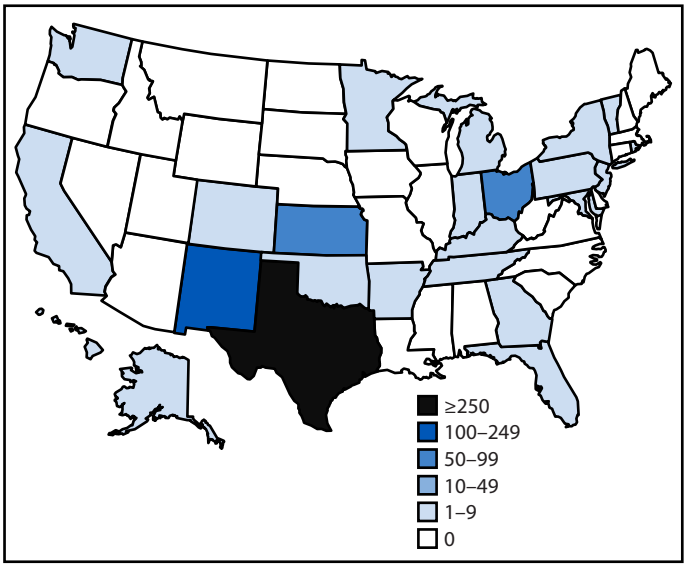
### Characteristics of Reported Measles Cases

During January 1–April 17, 2025, a total of 800 confirmed measles cases were reported in 25 U.S. jurisdictions (Figure 1). The highest number of weekly cases (99) was reported during the week ending March 22 (Figure 2). Median patient age was 9 years (IQR = 4–23 years); 249 (31%) patients were aged <5 years, 304 (38%) were aged 5–19 years, 231 (29%) were aged ≥20 years, and age was unknown for 16 (2%) patients (Table). Among all measles patients, 771 (96%) were unvaccinated or their vaccination status was unknown, 10 (1%) had received 1 dose of MMR vaccine, and 19 (2%) had received 2 doses. For Texas cases, it was not possible to disaggregate unvaccinated patients from those with unknown vaccination status because the Texas Immunization Registry requires explicit consent by law (i.e., opt-in) to enroll. Among 210 measles patients (excluding 590 cases reported by Texas), 162 (77%) were unvaccinated, six (3%) had received 1 dose of MMR vaccine, 12 (6%) had received 2 doses, and the vaccination status of 30 (14%) was unknown. Among all 800 cases, 790 (99%) occurred among U.S. residents. Overall, 85 (11%) patients were hospitalized; 56 (66%) of those were unvaccinated, one (1%) had received 1 dose of MMR vaccine, and the vaccination status of 28 (33%) was unknown. Three measles deaths were reported to CDC; two confirmed in Texas in unvaccinated school-aged children with no known underlying medical conditions, and one confirmed in New Mexico in an unvaccinated adult. Most cases (557; 70%) were laboratory-confirmed; among 251 (31%) cases from which specimens were available for molecular sequencing, all were confirmed as wild-type virus strain with 225 (90%) identified as genotype D8 and 26 (10%) as genotype B3.

### International Importations

Forty-eight (6%) cases were directly imported from other countries, including 44 (92%) among U.S. residents who had traveled abroad; 752 (94%) cases were U.S.-acquired. Fifteen (31%) importations resulted in secondary cases. Among the 48 internationally imported measles cases, 33 (69%) patients were unvaccinated, one (2%) had received 1 dose of MMR vaccine, four (8%) had received 2 doses, and the vaccination status of 10 (21%) patients was unknown. All 33 of the unvaccinated persons with imported measles were age-eligible for vaccination per ACIP, including 10 infant travelers aged 6–11 months. Source countries of the 48 imported measles cases included Canada (10 cases), Vietnam (10), Mexico (seven), Pakistan (three), the Philippines (two), Saudi Arabia (two), and one imported case each from Afghanistan, Australia, Guinea, Netherlands, Somalia, Spain, and Uganda; a source

**FIGURE 1. Reported number of confirmed* measles cases, by state (N = 800) — United States, January 1–April 17, 2025**




* An acute febrile rash illness with laboratory confirmation of measles or a direct epidemiologic link to a laboratory-confirmed measles case.

** Epidemiologic linkages include having known or suspected contact with an infectious measles patient during the exposure period (7–21 days before rash onset) and living in or visiting a geographic area with ongoing measles transmission during the exposure period.

†† Genotyping was performed at CDC and at the Vaccine Preventable Disease Reference Centers of the Association of Public Health Laboratories.

§§ 45 C.F.R. part 46.102(l)(2), 21 C.F.R. part 56; 42 U.S.C. Sect. 241(d); 5 U.S.C. Sect. 552a; 44 U.S.C. Sect. 3501 et seq.